

RESEARCH

Open Access



Integrative analysis identifies FBXO5 as a critical mediator of CRPC progression and bone metastatic potential

Rajnikant Raut¹, Devesh Srivastava¹, Amit Kumar Chakraborty², Manish V. Bais², Parul Mishra³ and Ashish Misra^{1*}

*Correspondence:

Ashish Misra

ashishmisra@bt.iith.ac.in

¹Department of Biotechnology,
Indian Institute of Technology
Hyderabad, Kandi,
Sangareddy 502285, India

²Department of Translational
Dental Medicine, Boston University
Henry M. Goldman School of
Dental Medicine, 700 Albany Street,
Boston, MA 02118, USA

³Department of Animal Biology,
School of Life Sciences, University
of Hyderabad, Hyderabad
500046, India

Abstract

Introduction Castration-resistant prostate cancer (CRPC) poses a major clinical challenge due to its aggressive nature, therapeutic resistance, and frequent progression to bone metastasis. Emerging evidence suggests that E3 ubiquitin ligases play a key role in cancer progression, yet their involvement in CRPC remains poorly understood.

Methods and Results In this study, we employed machine learning (ML) algorithms and integrative bioinformatics approaches to analyze the gene expression profiles of E3 ubiquitin ligases in CRPC patients. Our ML analysis identified FBXO5, TRIM52, PDZRN4, CCNF, TRIM59, and SHPRH as key predictor variables in CRPC. Furthermore, differential expression analysis revealed that FBXO5, TRIM52, CCNF, and PDZRN4 are significantly dysregulated in patient samples exhibiting strong concordance with the ML results. Notably, FBXO5 exhibits robust interactive, survival, and gene expression correlation with BARD1, BRCA1, UHRF1, and SKP2; key regulators of genomic instability and drug resistance in CRPC. Additionally, FBXO5 also exhibits correlative association with CXCR4, a key mediator of prostate cancer bone metastasis. To experimentally validate these in-silico results, we performed functional assays, including cell proliferation, wound healing, and flow cytometry-based cell cycle analysis in FBXO5-knockdown prostate cancer cells. Our results demonstrate that FBXO5 depletion significantly impairs cell proliferation and migration while inducing G1-phase arrest, supporting its oncogenic role in CRPC.

Conclusion Taken together, our findings establish FBXO5 as a critical mediator of CRPC progression and bone metastatic potential, underscoring its importance as a promising therapeutic target for this aggressive disease.

Keywords PCa, CRPC, Bone metastasis, TCGA, Machine learning, E3 ligase, FBXO5

1 Introduction

Castration-resistant prostate cancer (CRPC) presents a formidable clinical challenge due to its aggressive nature, limited treatment options and frequent progression to bone metastases. Despite initial responsiveness to androgen deprivation therapies (ADT), a significant proportion of patients eventually develop resistance, leading to disease



© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

progression and mortality typically within 18–24 months post-standard prostate cancer (PCa) treatment [1]. Current therapeutic options for CRPC, such as chemotherapy, exhibits modest success rates, necessitating the exploration of alternative approaches. With rapid improvements in the development of low toxicity regimens and efficient markers for tumor response, chemotherapy can now be augmented with other therapies to overcome resistance post-ADT failure [2–5].

Various studies have explored pathways associated with CRPC, allowing researchers to adopt treatment strategies beyond the traditional androgen receptor (AR)-targeted therapies. Earlier studies have highlighted the importance of novel drivers like IGF2BP2, CLK1, and YTHDC1, which promote CRPC progression by regulating the expression and stability of AR splice variant 7 (AR-V7) [6–8]. Recently, it has been demonstrated that Microtubule Affinity-Regulating Kinase 3. promotes CRPC progression by modulating CRPC-specific gene networks including those involved in androgen signaling, epithelial-to-mesenchymal transition (EMT), mTORC1, and MYC pathways [9, 10]. These findings support the rationale for combinatorial approaches targeting diverse pathways to further suppress AR signaling and enhance therapeutic efficacy [11]. In this context, investigating specific protein families crucial for cellular growth and development offers a novel approach to identifying potential drug targets. Among these, E3 ligases have garnered attention due to their involvement in various cellular processes. E3 ligases, in conjunction with ubiquitin activating enzyme E1 and ubiquitin conjugating enzyme E2, orchestrate a three-enzyme cascade crucial for catalyzing the ubiquitination process. This process involves the transfer of an ubiquitin moiety to a lysine site of the target substrate, thereby regulating cellular homeostasis [12]. Dysregulation of E3 ubiquitin ligases has been implicated in numerous cancers via diverse mechanisms, resulting in altered gene expression of downstream targets. This dysregulation contributes to genomic instability and uncontrolled proliferation, ultimately driving malignancy and therapy resistance. Specifically, E3 ligases regulate key cellular growth pathways, including MAPK or PI3K-AKT-mTOR, rendering them attractive targets for anticancer therapies. Moreover, they modulate cell stress response and maintain homeostasis between cellular growth and survival signals underscoring their significance in tumor initiation and progression [13]. Findings also suggest that E3 ligases are associated with the regulation of tumor bone metastasis [14].

Given the pivotal role of E3 ligases in cancer pathogenesis and metastasis, investigating their expression in CRPC patients holds promise for identifying novel therapeutic targets. In this study, we have leveraged machine learning and bioinformatics techniques to identify candidate E3 ligases associated with CRPC. Through supervised learning analyses of gene expression data from The Cancer Genome Atlas (TCGA), we identified dysregulated E3 ligases in CRPC patients. Subsequent analysis, including Kaplan–Meier survival estimation, correlation analysis, and hub gene detection using the MCC algorithm, led to the identification of FBXO5 (F-box only protein 5) as a potential oncogenic driver of CRPC progression and metastasis. FBXO5 is an E3 ubiquitin ligase, that is frequently overexpressed in multiple cancers and associated with poor prognosis, lymph node metastasis, and advanced tumor stage. [15]. It plays a critical role in cell cycle regulation by inhibiting the anaphase-promoting complex/cyclosome (APC/C), thereby promoting tumor cell proliferation, migration, and invasion [15–18]. It has also been implicated in the progression of breast cancer, glioma, and colorectal cancer;

however, its role in prostate cancer, particularly in CRPC and metastasis, has not yet been explored. To experimentally validate the role of FBXO5 in driving CRPC, we performed cell proliferation, migration, and cell cycle analysis experiments in cells treated with siRNA against FBXO5. Taken together, our results demonstrate the role of FBXO5 in CRPC progression and set the stage for future studies into its role as a therapeutic target.

2 Methodology

2.1 Data acquisition and processing

Gene expression data of CRPC patients was obtained from the online public repository. Total 52 healthy prostate samples were obtained from the TCGA database from the TCGA-PRAD study. TCGA-PRAD is a study that contains more than $n=498$ Prostate adenocarcinoma samples, however it does not contain CRPC samples. Therefore, we obtained around $n=208$ CRPC patient samples from PRAD-SU2C-2019 [19] and $n=99$ samples from WCDT-MCRPC project of TCGA. All the gene expression files were fetched and compiled using a Bioconductor package *TCGAbiolinks* (v2.22.0) in RStudio (v4.1.1) in December 2021. The gene expression values in these datasets were normalized to FPKM. In the obtained data, genes were labeled as Ensembl ids. We further converted all the Ensembl ids to a gene symbol using *bioMart* (v2.50.0) package (<https://bioconductor.org/packages/release/bioc/html/biomaRt.html>). The list of E3 ligases was obtained from the databases prepared by Barbara Medvar et al. [20]. Gene expression values of all the E3 ligase from the overall dataset were extracted using the R function *subset*.

To evaluate the potential batch-related variability, we applied the ComBat-seq algorithm [21] algorithm of *sva* R package to the combined gene expression dataset and performed Principal Component Analysis (PCA) before and after batch correction. The proportion of variance explained by the first two principal components changed only minimally (PC1: 12.6% to 12.8%; PC2: 10.1% to 9.7%), suggesting that batch effects did not substantially influence the overall variability. (Supplementary Fig. 1). Given the minimal impact of correction and our goal to preserve genuine biological signals, we opted to proceed with the uncorrected data for downstream analyses including Boruta feature selection, Random Forest modeling, and differential gene expression (DGE). Integrated gene expression dataset before and after batch correction are presented in Supplementary Table 1 and 2, respectively.

2.2 Machine learning for biological data

Random Forest and Boruta were the two supervised learning algorithms considered in this analysis [22–24]. Boruta specializes in feature selection, whereas Random Forest performs feature selection followed by predictive model building. Both *randomForest* and *Boruta* packages were imported from CRAN repository into the RStudio and analysis was performed. We split the data into 75% and 25% as training and test dataset, respectively. Both algorithms sort the top predictor variable based on an importance given to each predictor variable (here genes are the predictor variable). We selected the top 10 predictor variables from both algorithms for further analysis.

2.3 Differential expression analysis followed by KM survival estimation

Differential gene expression (DGE) analysis of the PRAD-SU2C-2019 and WCDT-MCRPC datasets was performed using healthy prostate samples from TCGA-PRAD as controls. The analysis was performed using the TCGAanalyze_DEA function within the limma pipeline in TCGAbiolinks package. E3 ubiquitin ligases were considered significant if they exhibited co-expression in the same direction across both the datasets and had an FDR < 0.05. FDR correction for multiple hypothesis testing was performed using the Benjamini–Hochberg procedure. Heatmaps were generated in RStudio using Bioconductor package ComplexHeatmap (<https://www.bioconductor.org/packages/release/bioc/html/ComplexHeatmap.html>) combined with Circlize (<https://cran.r-project.org/web/packages/circlize/index.html>).

Furthermore, Kaplan–Meier (KM) survival estimation was performed in cBioPortal using the PRAD-SU2C-2019 dataset. Patients were divided into high and low expression groups based on z-scores (Patients with negative z-scores were considered as low expression group and positive z-scores were considered as high expression group). Z-score is a statistical measure that quantifies the number of standard deviations a value (x) deviates from the population mean (μ), calculated as $z = (x - \mu) / \sigma$, where σ represents the standard deviation. This standardization allows robust comparison of gene expression across samples and facilitates survival analysis by providing a binary classification of patients. The data was plotted using GraphPad PRISM software (<https://www.graphpad.com>).

2.4 PPI and hub gene detection

In order to determine the relationship between common dysregulated E3 ligases we created the PPI network (protein–protein interaction network) using the STRING database [25]. Cytoscape software (v3.9.0) was used to import and create a model of the PPI network obtained from the STRING database [25, 26]. Furthermore, the *cytoHubba* plugin within Cytoscape was used to screen out the hub genes among the common dysregulated E3 ligases [27]. Hub genes screening was done using the MCC algorithm (Maximal Clique Centrality) in the *cytoHubba* plugin, as studies have indicated the MCC gives the most efficient and accurate hub genes [27]. Therefore, we considered genes with top 10 MCC scores as a hub gene (Supplementary Table 7 shows interaction scores and node size).

2.5 Cell culture

Prostate cancer cell lines 22Rv1 and PC3 cells were acquired from ATCC. Both cell lines were cultured in RPMI medium supplemented with 10% FBS and 1% Penicillin–Streptomycin cocktail. Cells were incubated in traditional growth conditions as 37 °C with 5% CO₂ and sub-cultured when they reached 70–80% confluence.

2.6 FBXO5 knockdown and quantitative real-time PCR

Human FBXO5 siRNA (Catalog No # sc-37611) and primers (Catalog No # sc-37611-PR) were purchased from Santa-Cruz Biotechnology and transfected into the cells using FuGENE® 6 Transfection Reagent. The nucleotide sequences are not available as they are the proprietary of Santa-Cruz Biotechnology. Real-time PCR (qRT-PCR) and

immunoblot analysis using FBXO5 antibody (Proteintech- Cat No. 10872–1-AP) was performed to confirm the successful FBXO5 knockdown.

2.7 Cell proliferation, migration, and cell cycle analysis

To perform the cell proliferation assays, cells were seeded into the 12-well plate and stained with the NucBlue™ Live ReadyProbes™ Reagent (Hoechst 33,342) (ThermoFisher Scientific: catalog#R37606) and imaged using Keyence microscope after 24 h. Quantification was performed using the in-built function of Keyence analyzer software. To perform the cell migration assay, cells were seeded into 12-well plates with a serum free medium and scratch was added using a sterile 1000ul pipette tip and imaged after 24 h for PC3 and 48/72 h for 22Rv1. To perform the cell cycle assay, cells were seeded into a 6-well plate for 24 h and harvested using routine trypsinization. Membrane permeabilization or fixation was performed using 70% ethanol following overnight incubation at 4 °C and stained with Propidium Iodide (PI) along with RNaseA and subjected to flow cytometry. Bar plots were plotted, and statistical analysis was performed using Graph-Pad PRISM software (<https://www.graphpad.com>).

3 Results

3.1 Supervised learning predicts top CRPC signatures within E3 ligase cohort

Random forest classifier is a supervised learning algorithm that makes decisions based on a labeled input data (training data) to identify the nature of sample data (test data) and detects the top predictor variables based upon importance given to each predictor variable [23]. Here, we considered CRPC data from project WCDT-MCRPC (n = 99) and PRAD-SU2C-2019 (n = 208) [19] along with 52 healthy samples (project: TCGA-PRAD) to run through a supervised learning algorithm. Initially, the gene expression matrix was labeled, row sampling was performed, and split into 75% and 25% for training and testing purposes, respectively. The model demonstrated exceptional performance, achieving perfect metrics across all evaluation parameters, including an AUC of 1.0, Precision of 1.0, Recall of 1.0, F1-score of 1.0, False Positive Rate (FPR) of 0, and False Negative Rate (FNR) of 0, using 100,000 decision trees (Supplementary Fig. 2). While these results are indicative of high accuracy, the possibility of overfitting due to the limited sample size in the dataset cannot be ruled out. This concern can only be adequately addressed through validation on larger, independent datasets, as demonstrated by Gaikwad et al., [28]. To visualize the candidates with the highest importance, we generated the variable importance plot. Among all the E3 ubiquitin ligases, SIAH1, TRIM52, ZNF131, PDZRN4, RBBP6, TRIM59, ZXDC, SHPRH, FBXO5, and CCNF were top 10 predictor variables as shown in a variable importance plot (Fig. 1A) (Supp. Table 3, 4).

Boruta is another supervised learning algorithm that works in line with the random forest algorithm for feature ranking and selection. The main advantage of using Boruta is that it clearly makes a decision if a variable is important or not and helps to screen variables that are statistically significant [24]. Following the pipeline and sample selection mentioned above for random forest, we performed the Boruta analysis with the maxRuns of 100. maxRuns is the number of times an algorithm is run which makes the algorithm more selective in picking the variables. We generated a variable importance plot for top 10 predictor variables. These include PDZRN4, SHPRH, TRIM59, RNFT2, FBXO5, ASB2, MAEA, TRIM52, CCNF, and UHRF1 as top contributors in prediction

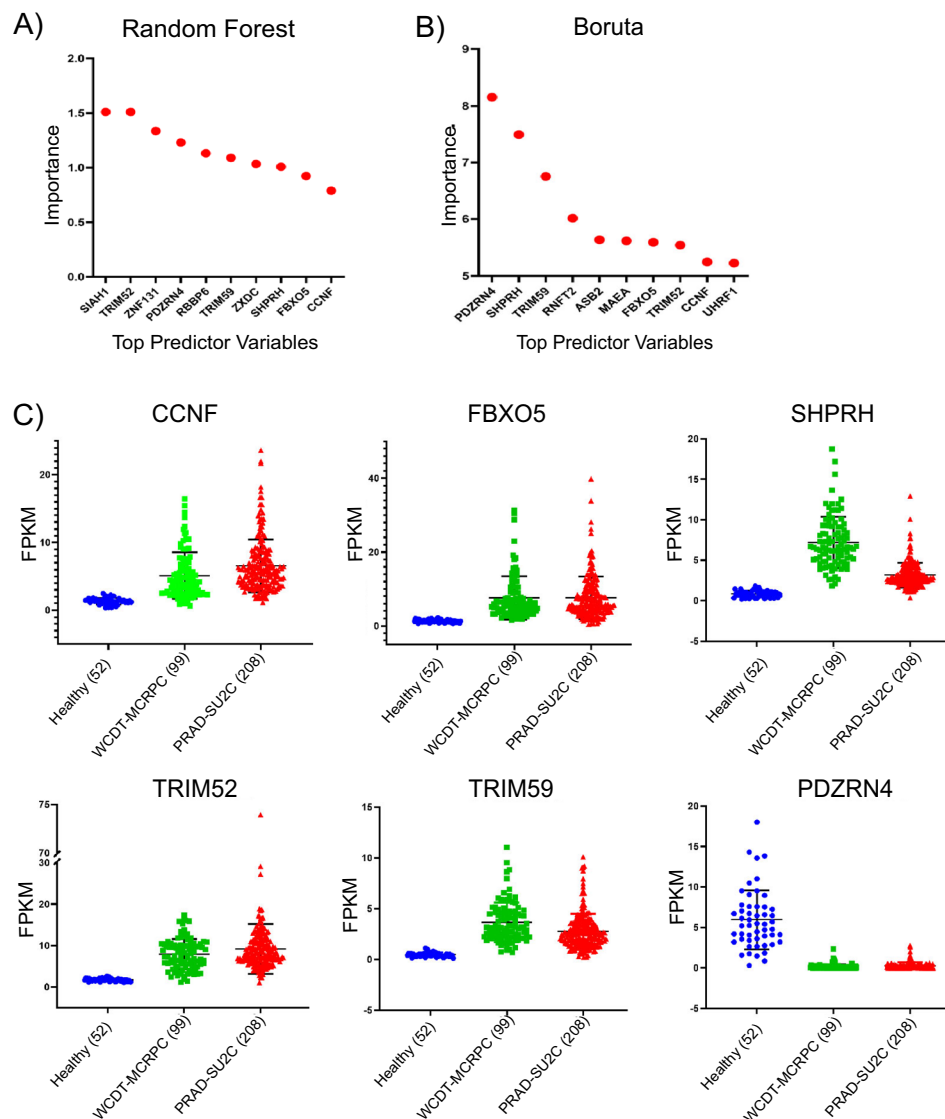


Fig. 1 Machine Learning Pipeline and Outputs. **A** Variable importance plot showing the top 10 CRPC predictors within E3 ligase cohort by random forest algorithm. **B** Variable importance plot showing the top 10 CRPC predictors within E3 ligase cohort by Boruta feature selection algorithm. **C** Expression plots representing mRNA expression of overlapping E3 ligases identified in CRPC patients

(Fig. 1B) (Supp. Figure 3, 4, & Supp. Table 5). Candidates, namely TRIM52, PDZRN4, TRIM59, SHPRH, FBXO5, and CCNF were common across both the analyses. Moreover, all the common candidates were highly upregulated in both CRPC data sets except PDZRN4, which is highly downregulated (Fig. 1C). This suggests that the predictions made by both the algorithms are irrespective of their gene expression status, but is associated with the consistent pattern detected in variables across the CRPC patients. Here, we conclude that both the supervised learning algorithms were efficient to predict important signatures associated with cancer progression in castration resistant prostate cancer patients.

3.2 Differential expression analysis followed by heatmap visualization of dysregulated E3 ligases in CRPC Patients.

Differential gene expression (DGE) analysis is a technique that considers the normalized gene expression value of a particular gene in two or more groups and interprets the differences in abundance of gene transcripts within a transcriptome [29]. Here, we performed DGE analysis of CRPC patients with two different data sets that are independent of each other and extracted the commonly dysregulated E3-ubiquitin ligases. Since two different data sets were considered, we performed their analysis separately with the *limma* pipeline. Around 120 candidates were differentially expressed in common, of which around 90 were upregulated and 30 were downregulated (Supp. Table 6). The top 10 upregulated genes were UHRF1, SIAH1, TRIM52, FBXO5, MEX3A, CCNE, TRIM25, ZNF131, and RNF34. Conversely, the top 10 downregulated genes were SOCS3, PDZRN4, ASB2, FBXO32, RFLP2, KCTD14, ASB5, RNF39, FBXO17, and LGALS3BP (Supp. Table 6). Because the expression of a gene can differ from patient to patient, heatmaps were generated to illustrate the expression of differentially expressed E3 ubiquitin ligases at the patient level. Notably, all the dysregulated E3 ligases followed the expression pattern reflected in DGE data, which is represented in a color scheme of high to low expression values along with their respective FDR and T statistics (Fig. 2A, B). Most importantly, the top dysregulated genes including TRIM52, FBXO5, CCNE, and PDZRN4, overlapped with the top predictor variables in our machine learning outcome and were present in a significant position in the volcano plot of differentially expressed E3 ubiquitin ligases (Fig. 2C, D). Interestingly, other than the overlapping genes, the remaining predictors were also amongst the top dysregulated genes in DGE analysis (Supp. Table 6). In summary, DGE analysis of both the datasets highlighted the top candidates and appeared in line with random forest and Boruta validating the significance of these genes in CRPC progression.

3.3 Survival estimation highlighted the strength of FBXO5

All the commonly dysregulated candidate genes were then subjected to Kaplan–Meier survival estimation to assess the effect of their dysregulation on CRPC patients' survival. Using the *z score* values of each gene, patients were divided into two groups: "High" (Positive *z score*) and "Low" (Negative *z score*) expression. Among 90 upregulated genes, FBXO5, G2E3, MNAT1, ZBTB33, and RNF34 were the only candidates (further termed as "Survival Signature Genes") that were found significantly associated with poor survival outcome in CRPC patients when upregulated (Fig. 3A–E). Conversely, RFLP2 was the only gene among 30 downregulated candidates that possessed a lower survival rate when downregulated (Fig. 3F). The Kaplan–Meier Survival estimation analysis of other genes that came out of our ML analysis is summarized in Supplementary Fig. 5. Interestingly, FBXO5 is the only gene among four candidates that overlapped with DGE and machine learning outcomes. In brief, our Kaplan–Meier Survival estimation highlights the potential of FBXO5 in CRPC.

3.4 FBXO5 network is associated with resistance in CRPC

Next, we asked if there is any relationship between FBXO5 and drug resistance in CRPC. We used the STRING database to create a PPI network (protein–protein interaction network) to discover the link between all the differentially expressed E3 ligases

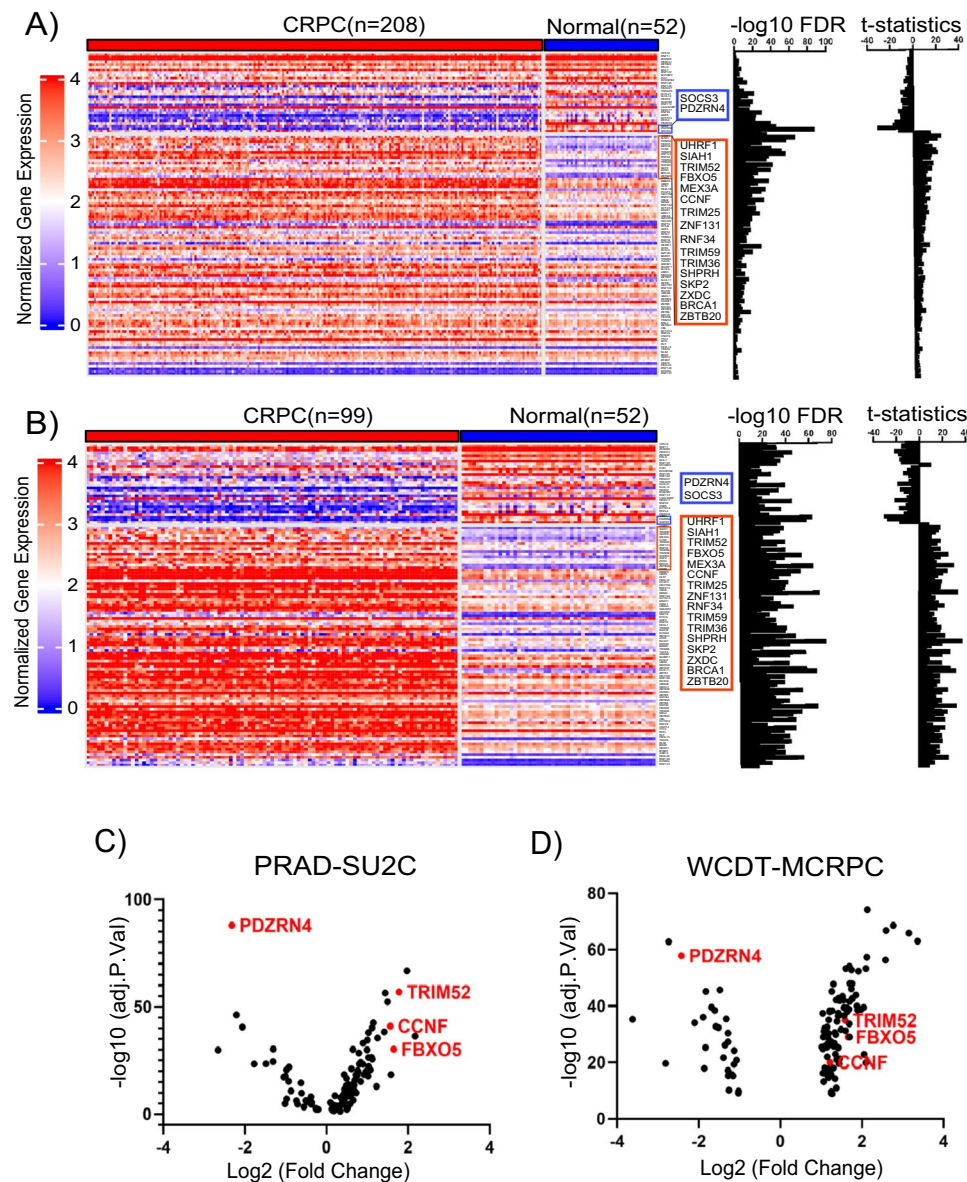


Fig. 2 DGE analysis and Heatmap Visualization of E3 ligase expression in CRPC patients. **A** Expression of E3 ligases in PRAD-SU2C-2019 patient data set; **B** Expression of E3 ligases in TCGA dataset. Red color shows the higher expression, and blue color shows lower expression, whereas white spaces represent intermediate expression. The upper panel represents downregulated, and the lower panel represents upregulated E3 ligases. Red rectangles indicate top E3 ligases that are overexpressed and blue rectangle indicates top downregulated E3 ligases. **C** Volcano plots of differentially expressed E3 ligases. E3 ligases overlapping with ML analysis are labelled. The false discovery rate or the adjusted p-values were calculated with Benjamini–Hochberg method

[25]. STRING database is a text mining based online tool which can help determine PPIs between desired proteins. The network was then imported to Cytoscape to create a model of hub gene network [27]. To screen the hub genes among the differentially expressed E3 ligases, we utilized the cytoHubba plugin in Cytoscape with the MCC algorithm (Maximal Clique Centrality), as studies have indicated the MCC gives the most efficient and accurate hub gene outcome. Here, we considered genes with top 10 MCC scores, which includes BARD1, BRCA1, UHRF1, SKP2, HUWE1, TRAF6, CBL, RAD18, ATRX, and FBXO5 (Please refer Supp. Table 7 for node size and string interaction score

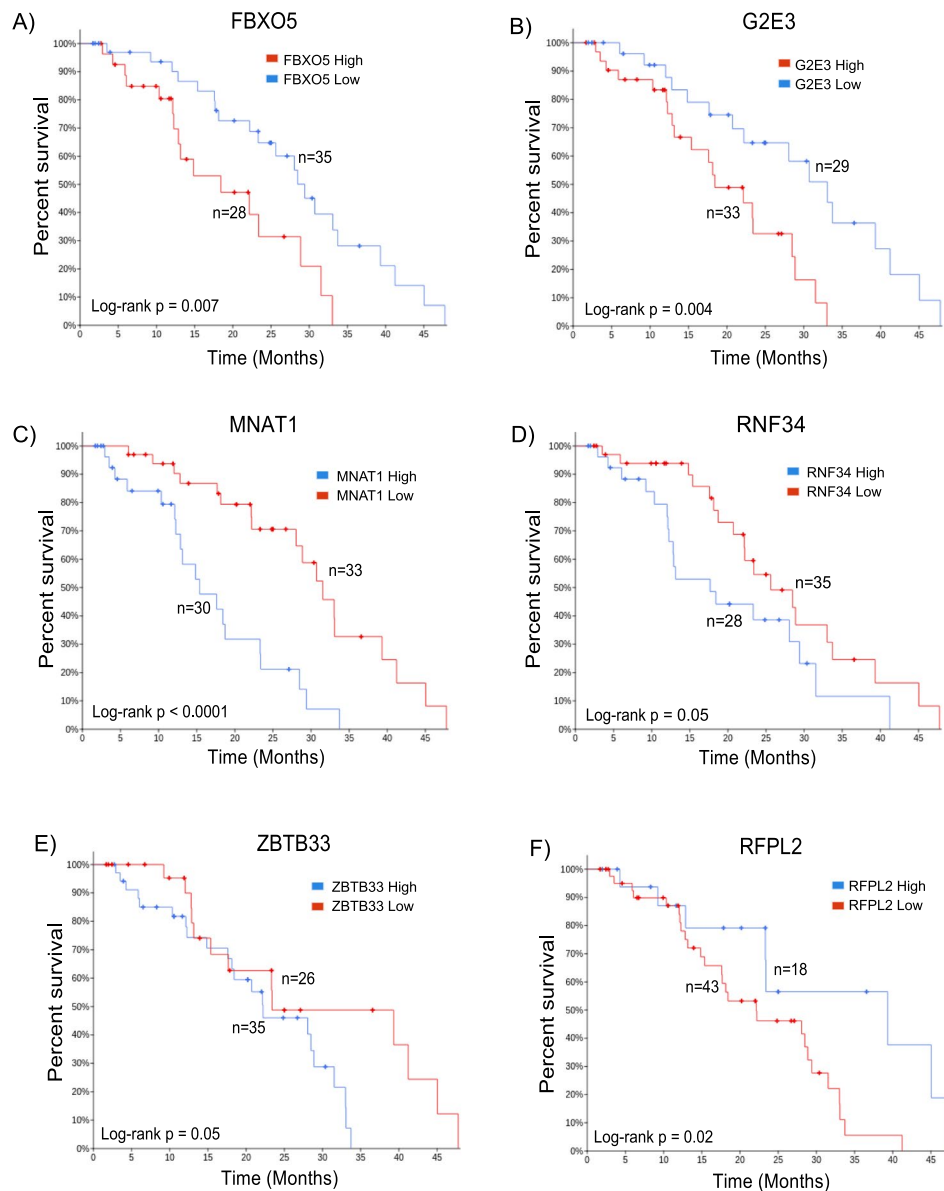


Fig. 3 Kaplan–Meier Survival estimation plot of dysregulated E3 ligases. p values were calculated using Log-Rank test

for each protein–protein interaction). Interesting, FBXO5 was one of the top 10 candidates and has an interactive relationship with prominent carcinogenic markers like BRCA1, BARD1, UHRF1, and SKP2 (Fig. 4A) (Supp. Figure 6) [30–33]. Studies have also shown that all four FBXO5 interactors are involved in the evolution of breast cancer, while UHRF1 is overexpressed in prostate cancer [34, 35], This highlights a substantial link between FBXO5 and hormone-related malignancies. Most importantly, SKP2 is known to contribute to drug resistance in CRPC. Next, we asked if the expression of FBXO5 correlates with these four carcinogenic markers. We performed correlation and KM survival analysis of FBXO5 and found that its expression is highly correlated with BRCA1, BARD1, UHRF1, and SKP2 (Fig. 4B). We also discovered that their individual co-upregulation with FBXO5 results in reduced overall survival in CRPC patients

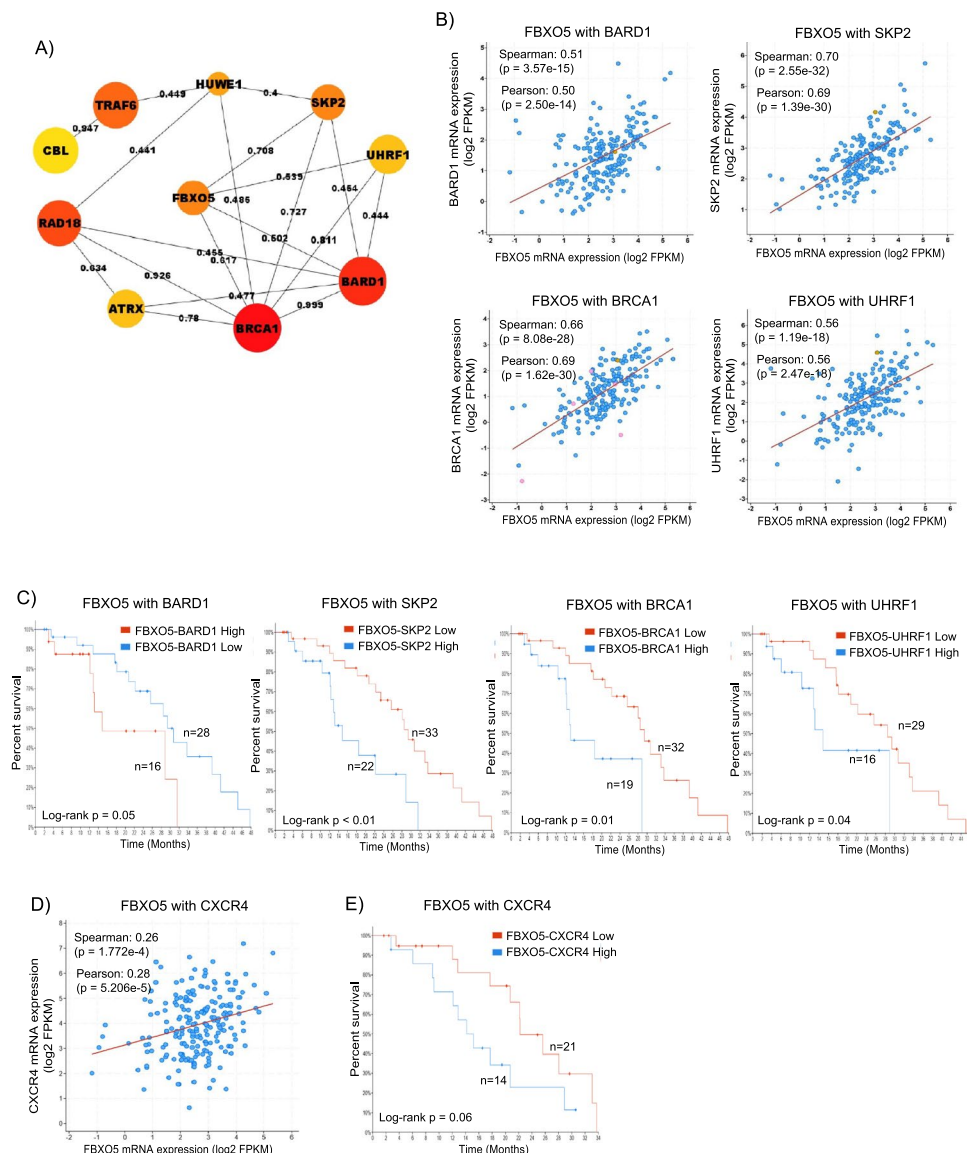


Fig. 4 FBXO5 network is associated with the resistance and metastasis in CRPC **A** Schematic representation of the hub gene interaction network constructed using CytoHubba plug in from Cytoscape. The size of the nodes and the color gradation are proportional to the STRING scores of respective interactions. The spectrum is such that the highest scoring gene is represented with a red node and the genes hence after are represented with a decreasing red and increasing yellow shade. The numbers on the edges correspond to the relative STRING score of the interaction. **B** Individual correlation of FBXO5 with BARD1, BRCA1, UHRF1, and SKP2. **C** KM survival analysis of BARD1, BRCA1, UHRF1, and SKP2 along with FBXO5. **D** Correlation analysis between FBXO5 and CXCR4

(Fig. 4C). Overall, this data suggests that FBXO5 is a prominent player in CRPC progression and could be associated with therapy resistance.

3.5 FBXO5 expression correlates with the bone metastasis in CRPC patients

Patients with PCa frequently develop bone metastases; up to 80% of patients with later-stage PCa and approximately 10% of patients diagnosed with primary disease have secondary skeletal tumors [36]. In bone metastases, CXCR4 is a potential therapeutic target due to its association with CXCL12 (SDF-1), which enhances tumor homing, survival, and treatment resistance [16]. Numerous CXCR4 antagonists and inhibitors are being

investigated to enhance therapeutic outcomes in PCa bone metastasis [37]. Interestingly, FBXO5 showed a significantly positive correlation with CXCR4 expression in CRPC patients (Fig. 4D) and also results in worse survival outcome when both are co-upregulated (Fig. 4E). These findings suggest that FBXO5 is a critical regulator of CRPC bone metastasis, promoting tumor cell migration and survival through CXCR4 activation.

3.6 Proliferation and migration assays validate the involvement of FBXO5 in malignant growth and metastasis

Proliferation is one of the key characteristics of cancer cells. To investigate the functional role of FBXO5 in promoting tumor cell proliferation, we silenced its expression using small-interfering RNA in 22Rv1 and PC3 cells and confirmed the knockdown efficiency using quantitative real-time PCR and immunoblot analysis (Fig. 5A and Supplementary

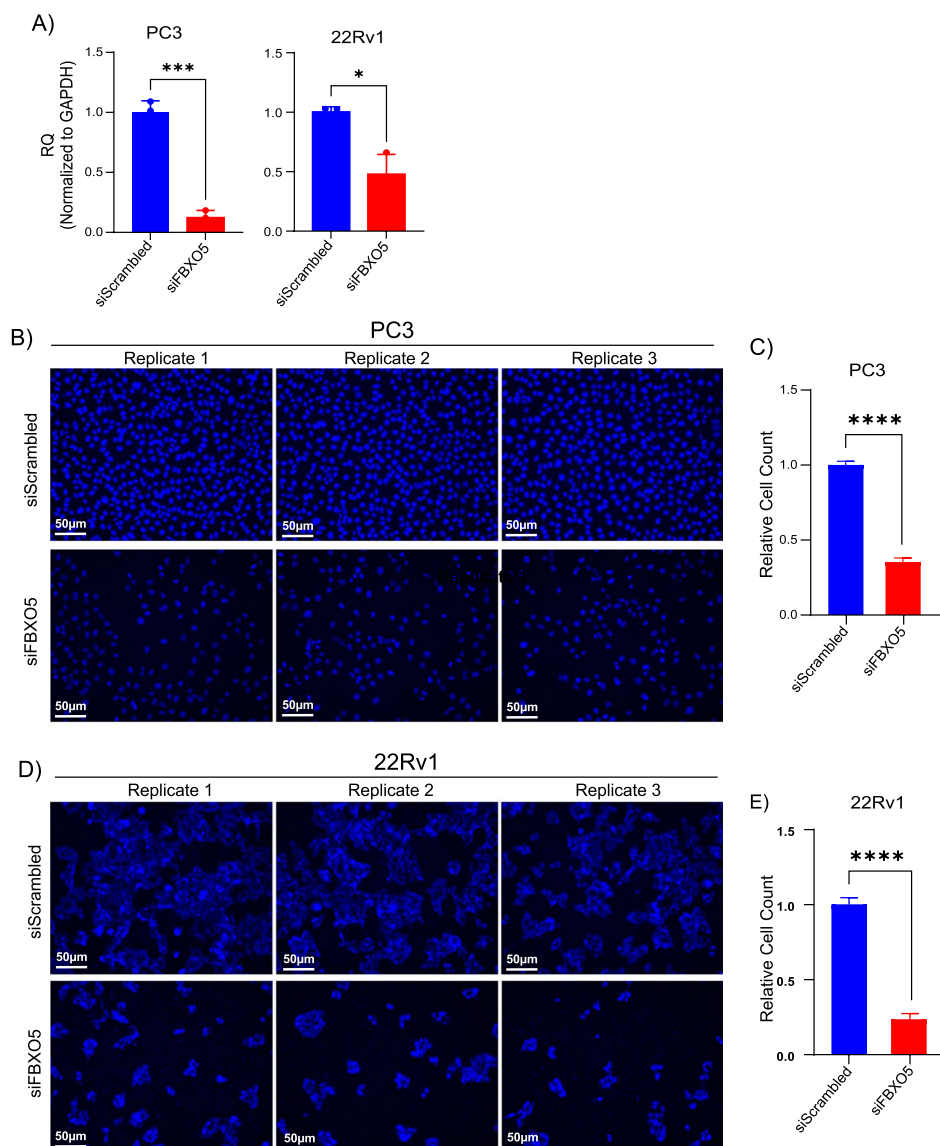


Fig. 5 FBXO5 promotes proliferation and migration. **A** Validation of siRNA mediated FBXO5 knockdown using qPCR. **B–E** Fluorescence microscopy imaging and quantification of proliferation in PC3 and 22Rv1 cells with FBXO5 knockdown. Students t-test was performed to calculate the p-values. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$

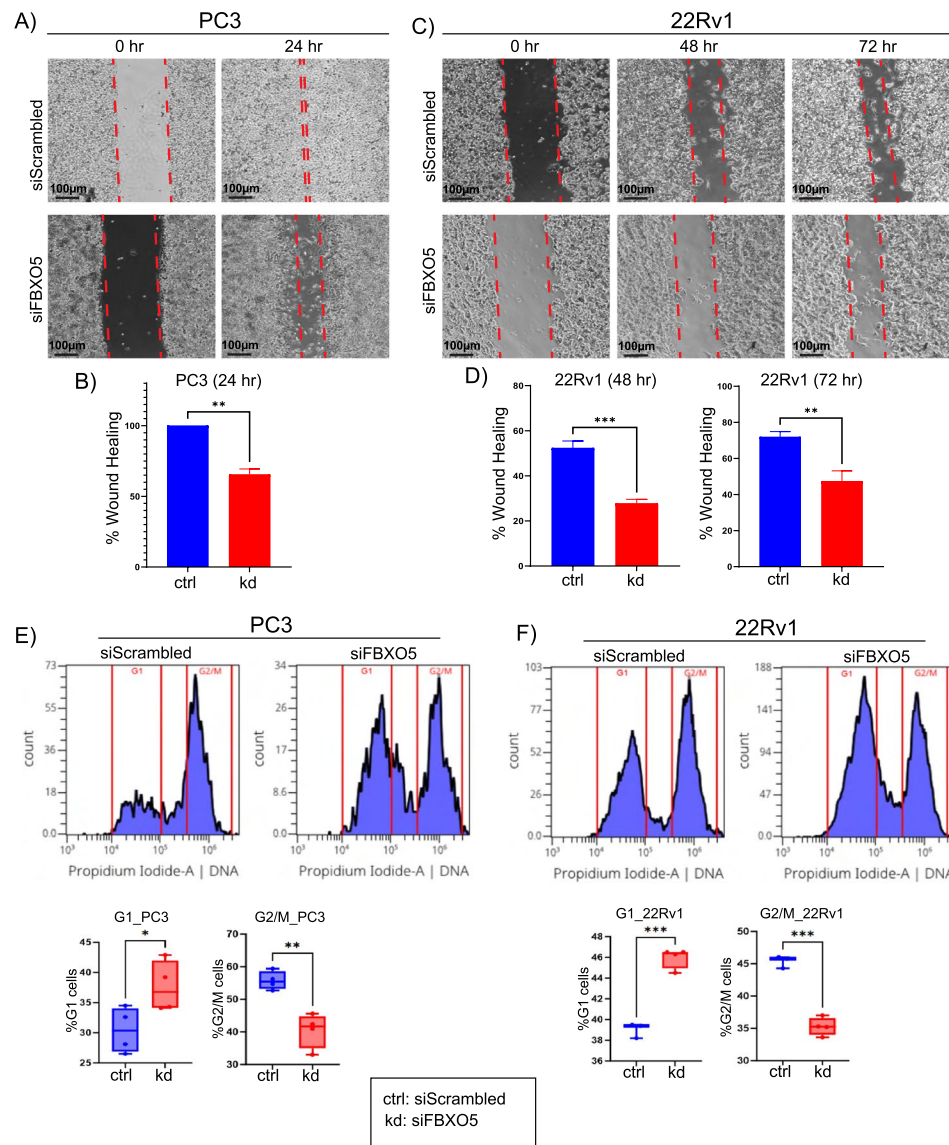


Fig. 6 A–D Bright-field images of wound healing of 22Rv1 and PC3 cells with FBXO5 knockdown. E–F Flow cytometry-based cell cycle stages quantification of 22Rv1 and PC3 cells with FBXO5 knockdown. Students t-test was performed to calculate the p-values. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$

Fig. 7A). Our results demonstrate that depletion of FBXO5 impairs the proliferation of both 22Rv1 and PC3 cells, suggesting its key role in cancer progression (Fig. 5B–E). In addition to proliferation, cell migration is a fundamental feature of metastatic cancer cells, particularly in the context of prostate cancer's propensity for bone metastasis. To assess whether FBXO5 influences the migration potential of cells, we performed wound healing assays following siRNA-mediated knockdown in 22Rv1 and PC3 cells. Our results demonstrate a significant reduction in migratory capacity in both 22Rv1 and PC3 cells upon FBXO5 silencing, with a more pronounced effect observed in 22Rv1 cells (Fig. 6A, B; Supplementary Figs. 7B, C). To evaluate the broader relevance of FBXO5 in prostate cancer cell viability, we analyzed gene dependency data from the DepMap portal (DEMETER v6). Consistent with our experimental results, the analysis revealed that

FBXO5 loss significantly reduces the fitness of multiple prostate cancer cell lines, further supporting its functional importance (Supplementary Fig. 7D).

Collectively, these results demonstrate that FBXO5 is an important driver for migration and proliferation in CRPC.

3.7 FBXO5 depletion induces cell cycle arrest

FBXO5, a member of the F-box protein family, regulates cell cycle progression by modulating the activity of anaphase-promoting complex (APC). To elucidate its role in cell cycle regulation, we performed propidium iodide (PI) based flow cytometry analysis in 22Rv1 and PC3 cells following siRNA-mediated knockdown of FBXO5. Our results revealed that FBXO5 depletion results in cell cycle arrest in the G1-phase in both cell lines (Fig. 6E, F) supporting the previously described function of FBXO5 in cell cycle regulation [38]. The reduced ability of cells to transition into the S-phase and G2/M phase upon FBXO5 knockdown underscores its crucial role in modulating cell cycle progression.

4 Discussion

CRPC continues to represent a critical challenge in male mortality, necessitating a deeper comprehension of its mechanisms to develop effective treatments. While previous studies have shed light on CRPC development and drug resistance mechanisms, our understanding remains limited, particularly concerning the constitutive expression of androgen receptors and associated cellular processes. To address this gap, we conducted a systematic study of E3-ubiquitin ligases utilizing machine learning algorithms, bioinformatics and experimental validation. These included feature selection, differential gene expression analysis, Kaplan–Meier survival estimation, hub gene detection, gene expression correlation, and experimental validation through proliferation assay, wound healing assay for cell migration, and cell cycle analysis using flow cytometry.

Our study employed supervised learning algorithms, including random forest and boruta, to identify key predictors among E3 ligases. Our analysis revealed TRIM52, PDZRN4, TRIM59, SHPRH, FBXO5, and CCNF as significant players in CRPC. Further differential gene expression analysis of CRPC patient samples identified 120 dysregulated genes, with FBXO5, TRIM52, PDZRN4, and CCNF emerging as the top candidates. These findings not only corroborated our machine learning predictions but also highlighted the importance of these genes in the pathophysiology of CRPC. Moreover, survival analysis of these dysregulated genes demonstrated that six of them, including FBXO5, were significantly associated with poor survival outcomes in CRPC patients, suggesting that FBXO5 may serve as a critical biomarker for CRPC prognosis. We employed supervised learning algorithms random forest and boruta to identify common predictor variables among E3 ligases. Our analysis revealed TRIM52, PDZRN4, TRIM59, SHPRH, FBXO5, and CCNF as significant predictors of CRPC. Further, differential gene expression analysis of CRPC patient samples identified 120 genes, with TRIM52, FBXO5, CCNF, and PDZRN4 emerging as the top dysregulated genes, aligning with the findings from machine learning analysis. This consistency underscored the significance of these genes in the pathophysiology of CRPC. Furthermore, survival estimation of dysregulated E3 ubiquitin ligases highlighted six genes associated with poor survival outcomes in CRPC patients, with FBXO5 standing out as the sole candidate consistently detected

across all the analyses. This pointed towards the potential of FBXO5 as a biomarker for CRPC prognosis and its importance in the E3-Ubiquitin ligase cohort. Furthermore, siRNA mediated knockdown of FBXO5 in 22Rv1 and PC3 cell lines leads reduced proliferation and migration, along with G1-stage cell cycle arrest. These findings provide functional evidence supporting FBXO5's involvement in CRPC progression.

FBXO5 (also known as EMI1) is a well-established regulator of anaphase-promoting complex/cyclosome (APC/C), modulating its activity during cell cycle. [39–43]. It acts both as a substrate and inhibitor of the APC-FZR1 complex, playing a dual role in G1-S transition and maintaining mitotic fidelity. [38]. These results perfectly align with our findings where FBXO5 depletion induces G1-phase cell cycle arrest. FBXO5 has emerged as a key regulator in multiple malignancies, including breast, ovarian, lung, and liver cancers, where its overexpression is frequently associated with increased tumor aggressiveness, genomic instability, and poor clinical outcomes. Mechanistically, FBXO5 overexpression promotes cellular proliferation, tetraploidy, and chromosomal instability, particularly in the context of p53 deficiency, suggesting a mutually exclusive relationship between FBXO5 activity and functional p53. Furthermore, FBXO5 has been shown to modulate the response to chemotherapeutic agents such as doxorubicin, selectively enhancing drug sensitivity in cancer cells while sparing normal cells, thereby underscoring its potential as a therapeutic target [44–46].

In our study, we also observed a strong gene expression correlation between FBXO5 and SKP2, another oncogenic E3 ligase involved in CRPC drug resistance through mechanisms such as epithelial-to-mesenchymal transition (EMT) and cancer stemness. Previous studies in chronic myeloid leukemia (CML) models have demonstrated that BCR-ABL stabilizes FBXO5, impairing SKP2 degradation and promoting oncogenic signaling. Our findings suggest that the FBXO5-SKP2 axis may contribute to therapeutic resistance in CRPC, underscoring the potential interplay between these two E3 ligases in driving disease progression [33, 47–49]

Additionally, the therapeutic relevance of FBXO5 is further supported by its post-transcriptional regulation by METTL16, an m6A methyltransferase that enhances its mRNA stability and promotes metastatic progression in breast cancer [50]. Interestingly, the expression of FBXO5 is capable of altering the tumor microenvironment where it positively associates with the levels of infiltrating regulatory T cells (Tregs) in most cancers but has a negative correlation with CD8+ T cells, NK/NKT cells, and Th2 cells [51]. This suggests that FBXO5 may play a role in modulating anti-tumor immune responses, potentially impacting the immune landscape of CRPC. While, direct inhibitors of FBXO5 are currently unavailable likely due to the absence of detailed structural information, agents such as thapsigargin, which induce endoplasmic reticulum (ER) stress, have been shown to downregulate FBXO5 expression [52].

Moreover, the significant positive correlation between FBXO5 and CXCR4, a key player in PCa bone metastasis, highlights the importance of FBXO5 in driving metastasis in CRPC patients. The potential of this unexplored correlative FBXO5-CXCR4 axis is supported by the poor survival outcome in the patients with higher co-expression, suggesting the existence of a cooperative oncogenic pathway that may serve as a viable target for therapeutic intervention.

5 Limitations of the study

While our study provides compelling evidence for the role of FBXO5 in CRPC progression and its potential therapeutic relevance, several aspects require further investigation to enhance the translational impact of our results. First, due to the lack of matched normal tissue within CRPC, we utilized healthy prostate samples from TCGA-PRAD for comparison of differential gene expression. Although, our batch effect assessment using the ComBat algorithm showed minimal variance that can be attributed to dataset differences, we opted to use uncorrected data to preserve biological signals relevant to downstream analyses. Nonetheless, the integration of multi-cohort data may introduce some degree of heterogeneity which should be considered while interpreting the results. Secondly, we performed functional validation of FBXO5 using two commonly studied prostate cancer cell lines, 22Rv1 and PC3. While these cell line models effectively represent aspects of CRPC biology, they may not fully capture the heterogeneity present in patient tumors. Future studies involving a broader panel of CRPC models, including patient-derived organoids and xenografts, will be able to provide a more comprehensive understanding of the role of FBXO5 in CRPC progression.

6 Conclusion

In conclusion, our study provides comprehensive insights into the involvement of FBXO5 in CRPC progression and bone metastasis. Our consistent findings across computational analysis and experimental validations underscore FBXO5's significance as a potential therapeutic target in CRPC treatment. However, further research focusing on FBXO5 inhibition in in vivo clinical models could pave the way for novel therapeutic interventions to combat CRPC and improve patient outcomes.

Abbreviations

CRPC	Castration resistant prostate cancer
PcA	Prostate cancer
ADT	Androgen deprivation therapy
AR	Androgen receptor
MCC	Maximal clique centrality
TCGA	The Cancer Genome Atlas
STRING	Search tool for the retrieval of interacting genes or proteins
ML	Machine learning
DGE	Differential gene expression
KM	Kaplan–Meier
FDR	False discovery rate
DEGs	Differentially expressed genes
ROC	Receiver operating characteristics curve
AUC	Area under the ROC curve
FPR	False positive rate
FNR	False negative rate
PI	Propidium iodide

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1007/s12672-025-03069-y>.

Supplementary material 1. Principal component analysis (PCA) plots before and after the batch effect removal

Supplementary material 2. ROC plot for the accuracy of random forest model

Supplementary material 3. Overall variable importance plot with feature selection algorithm Boruta. Here, shadow attributes are represented by blue boxes, significant attributes are represented by green boxes, tentative attributes are represented by yellow boxes, and unimportant attributes are represented by red boxes.

Supplementary material 4. History of variable importance. In essence, alternative Boruta object plot method, displaying a matplot of attribute importance with run. Here, shadow attributes are represented by blue lines, significant attributes are represented by green lines, tentative attributes are represented by yellow lines, and unimportant

attributes are represented by red lines.

Supplementary material 5. KM survival plots for top ML output gene showing statistically insignificant survival outcome.

Supplementary material 6. Overall protein-protein interaction network of dysregulated E3 ligases using STRING database.

Supplementary material 7. Western blot analysis of FBXO5 expression in 22Rv1 and PC3 cells following transfection with siScrambled (control) or siFBXO5. ACTB (β -actin) was used as a loading control. Bright-field microscopy images of wound healing assay in B) PC3 and C) 22Rv1 cell lines transfected with siScrambled or siFBXO5. Images were recorded at indicated time points. (D) Fitness dependency analysis showing reduced fitness of prostate cancer cell lines upon FBXO5 depletion.

Supplementary material 8.

Acknowledgements

This work is supported by SERB (CRG/2022/008672) and ICMR (NCD/Ad-hoc/224/2022-23) grant to A.M. P.M. acknowledges funding support from DBT BUILDER (BT/INF/22/SP41176/2020), MoE-STARs/STARs-1/PID (STARs1/634) and UoH-IoE-RC5-22-011 grants. P.M. is a recipient of ICMR-DHR Fellowship from ICMR, Gol. R.R. duly acknowledges CSIR-SRF fellowship support. D.S. is a recipient of PMRF.

Author contributions

R.R. P.M. and A.M. conceived and designed the experiments. R.R. designed and performed bioinformatics analysis as well as experimental validation with help from D.S. and A.C. R.R., D.S., A.C., P.M., M.B., and A.M. analyzed and interpreted the data and wrote the manuscript. All authors reviewed the paper and provided comments.

Funding

This work is supported by SERB (CRG/2022/008672) and ICMR (NCD/Ad-hoc/224/2022-23) grant to A.M. P.M. acknowledges funding support from DBT BUILDER (BT/INF/22/SP41176/2020), MoE-STARs/STARs-1/PID (STARs1/634) and UoH-IoE-RC5-22-011 grants. P.M. is a recipient of ICMR-DHR Fellowship from ICMR, Gol. R.R. duly acknowledges CSIR-SRF fellowship support. D.S. is a recipient of PMRF.

Availability of data and materials

Gene expression data of CRPC patients analyzed in this study was acquired from the GDC portal (Genomic Data Commons data portal) using following project IDs; Healthy controls: TCGA-PRAD (<https://portal.gdc.cancer.gov/projects/TCGA-PRAD>); CRPC: WCDT-MCRPC (<https://portal.gdc.cancer.gov/projects/WCDT-MCRPC>) and PRAD_SU2C_2019 (<https://pubmed.ncbi.nlm.nih.gov/31061129/>) of The Cancer Genome Atlas (TCGA). All the gene expression data files are publicly available.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Received: 19 March 2025 / Accepted: 24 June 2025

Published online: 07 August 2025

References

- Cong X, He Y, Wu H, Wang D, Liu Y, Shao T, Liu M, Yi Z, Zheng J, Peng S, Ding T. Regression of castration-resistant prostate cancer by a novel compound HG122. *Front Oncol*. 2021;11:650919. <https://doi.org/10.3389/fonc.2021.650919>.
- Esmali B, Hidaji L, Adinin RB, Faustina M, Coats C, Arbuckle R, Rivera E, Valero V, Tu S-M, Ahmadi MA. Blockage of the lacrimal drainage apparatus as a side effect of docetaxel therapy. *Cancer*. 2003;98:504–7. <https://doi.org/10.1002/cncr.11527>.
- Hussain M, Tangen CM, Lara PN, Vaishampayan UN, Petrylak DP, Colevas AD, Sakr WA, Crawford ED. Southwest Oncology Group, Ixabepilone (epothilone B analogue BMS-247550) is active in chemotherapy-naïve patients with hormone-refractory prostate cancer: a southwest oncology group trial S0111. *J Clin Oncol*. 2005;23:8724–9. <https://doi.org/10.1200/JCO.2005.02.4448>.
- Hussain A, DiPaola RS, Baron AD, Higano CS, Tchekmedyian NS, Johri AR. Phase II trial of weekly patupilone in patients with castration-resistant prostate cancer. *Ann Oncol*. 2009;20:492–7. <https://doi.org/10.1093/annonc/mdn665>.
- de Bono JS, Oudard S, Ozguroglu M, Hansen S, Machiels J-P, Kocak I, Gravis G, Bodrogi I, Mackenzie MJ, Shen L, Roessner M, Gupta S, Sartor AO. TROPIC Investigators. Prednisone plus cabazitaxel or mitoxantrone for metastatic castration-resistant prostate cancer progressing after docetaxel treatment: a randomised open-label trial. *Lancet*. 2010;376:1147–54. [https://doi.org/10.1016/S0140-6736\(10\)61389-X](https://doi.org/10.1016/S0140-6736(10)61389-X).
- Saini T, Srivastava D, Raut R, Mishra P, Misra A. Insulin-like growth factor 2 mRNA-binding protein 2 (IGF2BP2) promotes castration-resistant prostate cancer progression by regulating AR-V7 mRNA stability. *Cancer Rep*. 2025;8:e70096. <https://doi.org/10.1002/cnr2.70096>.

7. Saini T, Gupta P, Raut R, Nayak V, Bharathnaveen P, Mishra P, Misra A. AR-V7 expression facilitates accelerated G2/M phase transition in castration-resistant prostate cancer. *Exp Cell Res*. 2024;438: 114026. <https://doi.org/10.1016/j.yexcr.2024.114026>.
8. Gupta P, Srivastava D, Nayak V, Rao BV, Suseela K, Sharma R, Rajappa SJ, Rao TS, Mishra P, Misra A. YTHDC1 orchestrates oncogenic splicing via the CLK1-SRSF1 splicing machinery to regulate castration-resistant prostate cancer progression. *Mol Cell Biochem*. 2025. <https://doi.org/10.1007/s11010-025-05302-2>.
9. Raut, P. Gupta, T. Saini, P. Mishra, A. Misra, Src kinase: An attractive therapeutic target for prostate cancer, in: Protein Kinase Inhibitors, Elsevier, 2022; pp. 479–503. <https://doi.org/10.1016/B978-0-323-91287-7.00029-6>.
10. Raut R, Srivastava D, Nayak V, Saini T, Gupta P, Chakraborty AK, Choudhury C, Bais MV, Mishra P, Misra A. Clinical data investigation identifies MARK3 as an oncogenic driver in castration-resistant prostate cancer. *Biochem Biophys Res*. 2025;42: 102003. <https://doi.org/10.1016/j.bbrep.2025.102003>.
11. Shore ND, Morgans AK, Ryan CJ. Resetting the bar of castration resistance - understanding androgen dynamics in therapy resistance and treatment choice in prostate cancer. *Clin Genitourin Cancer*. 2021;19:199–207. <https://doi.org/10.1016/j.clg.2020.08.008>.
12. Yang Q, Zhao J, Chen D, Wang Y. E3 ubiquitin ligases: styles, structures and functions. *Mol Biomed*. 2021;2:23. <https://doi.org/10.1186/s43556-021-00043-2>.
13. Senft D, Qi J, Ronai ZA. Ubiquitin ligases in oncogenic transformation and cancer therapy. *Nat Rev Cancer*. 2018;18:69–88. <https://doi.org/10.1038/nrc.2017.105>.
14. Dong Y, Chen Y, Ma G, Cao H. The role of E3 ubiquitin ligases in bone homeostasis and related diseases. *Acta Pharm Sin B*. 2023;13:3963–87. <https://doi.org/10.1016/j.japsb.2023.06.016>.
15. Gao J, Yang D, Cao R, Huang H, Ma J, Wang Z, Xia J, Pan X. The role of Fbxo5 in the development of human malignant tumors. *Am J Cancer Res*. 2022;12:1456–64.
16. Xu Y, Zhang G, Liu Y, Liu Y, Tian A, Che J, Zhang Z. Molecular mechanisms and targeted therapy for the metastasis of prostate cancer to the bones (Review). *Int J Oncol*. 2024;65:104. <https://doi.org/10.3892/ijo.2024.5692>.
17. Zhang J, Zhang G, Wang K, Cui F, Yang H, Jiao Z. Exploring the role of FBXO5 in gastric cancer. *Mol Cell Probes*. 2023;69: 101915. <https://doi.org/10.1016/j.mcp.2023.101915>.
18. Jiang S, Zheng J, Cui Z, Li Y, Wu Q, Cai X, Zheng C, Sun Y. FBXO5 acts as a novel prognostic biomarker for patients with cervical cancer. *Front Cell Dev Biol*. 2023;11:1200197. <https://doi.org/10.3389/fcell.2023.1200197>.
19. Abida W, Cyrta J, Heller G, Prandi D, Armenia J, Coleman I, Cieslik M, Benelli M, Robinson D, Van Allen EM, Sboner A, Fedrizzi T, Mosquera JM, Robinson BD, De Sarkar N, Kunju LP, Tomlins S, Wu YM, Nava Rodrigues D, Loda M, Gopalan A, Reuter VE, Pritchard CC, Mateo J, Bianchini D, Miranda S, Carreira S, Rescigno P, Filipenko J, Vinson J, Montgomery RB, Beltran H, Heath EI, Scher HI, Kantoff PW, Taplin M-E, Schultz N, deBono JS, Demichelis F, Nelson PS, Rubin MA, Chinnaiyan AM, C.L. Sawyers, genomic correlates of clinical outcome in advanced prostate cancer. *Proc Natl Acad Sci USA*. 2019;116:11428–36.
20. Medvar B, Baghuram V, Pisitkun T, Sarkar A, Knepper MA. Comprehensive database of human E3 ubiquitin ligases: application to aquaporin-2 regulation. *Physiol Genom*. 2016;48:502–12. <https://doi.org/10.1152/physiolgenomics.00031.2016>.
21. Y. Zhang, G. Parmigiani, W.E. Johnson, ComBat-seq: batch effect adjustment for RNA-seq count data, *NAR Genom Bioinform* 2 lqaa078. <https://doi.org/10.1093/nargab/lqaa078>.
22. Acharjee A, Larkman J, Xu Y, Cardoso VR, Gkoutos GV. A random forest based biomarker discovery and power analysis framework for diagnostics research. *BMC Med Genomics*. 2020;13:178. <https://doi.org/10.1186/s12920-020-00826-6>.
23. Chen X, Ishwaran H. Random forests for genomic data analysis. *Genomics*. 2012;99:323–9. <https://doi.org/10.1016/j.jygeno.2012.04.003>.
24. Kursa MB, Rudnicki WR. Feature Selection with the Boruta Package. *J Soft Stat*. 2010. <https://doi.org/10.18637/jss.v036.i11>.
25. Szklarczyk D, Gable AL, Lyon D, Junge A, Wyder S, Huerta-Cepas J, Simonovic M, Doncheva NT, Morris JH, Bork P, Jensen LJ, von Mering C. STRING v11: protein–protein association networks with increased coverage, supporting functional discovery in genome-wide experimental datasets. *Nucleic Acids Res*. 2019;47:D607–13. <https://doi.org/10.1093/nar/gky1131>.
26. Shannon P, Markiel A, Ozier O, Baliga NS, Wang JT, Ramage D, Amin N, Schwikowski B, Ideker T. Cytoscape: a software environment for integrated models of biomolecular interaction networks. *Genome Res*. 2003;13:2498–504. <https://doi.org/10.1101/gr.1239303>.
27. Chin C-H, Chen S-H, Wu H-H, Ho C-W, Ko M-T, Lin C-Y. cytoHubba: identifying hub objects and sub-networks from complex interactome. *BMC Syst Biol*. 2014;8(Suppl 4):S11. <https://doi.org/10.1186/1752-0509-8-S4-S11>.
28. Gaikwad B, Patra A, Crawford CR, Miller EL. Self-supervised anomaly detection and localization for X-ray cargo images: generalization to novel anomalies. *Eng Appl Artif Intell*. 2025;140: 109675. <https://doi.org/10.1016/j.engappai.2024.109675>.
29. Conesa A, Madrigal P, Tarazona S, Gomez-Cabrero D, Cervera A, McPherson A, Szczesniak MW, Gaffney DJ, Elo LL, Zhang X, Mortazavi A. A survey of best practices for RNA-seq data analysis. *Genome Biol*. 2016;17:13. <https://doi.org/10.1186/s13059-016-0881-8>.
30. Nyberg T, Tischkowitz M, Antoniou AC. BRCA1 and BRCA2 pathogenic variants and prostate cancer risk: systematic review and meta-analysis. *Br J Cancer*. 2022;126:1067–81. <https://doi.org/10.1038/s41416-021-01675-5>.
31. Castro E, Eeles R. The role of BRCA1 and BRCA2 in prostate cancer. *Asian J Androl*. 2012;14:409–14. <https://doi.org/10.1038/aja.2011.150>.
32. Śniadecki M, Brzeziński M, Darecka K, Klasa-Mazurkiewicz D, Poniewierza P, Krzeszowiec M, Kmieć N, Wydra D. BARD1 and breast cancer: the possibility of creating screening tests and new preventive and therapeutic pathways for predisposed women. *Genes*. 2020;11:1251. <https://doi.org/10.3390/genes11111251>.
33. Ruan D, He J, Li C-F, Lee H-J, Liu J, Lin H-K, Chan C-H. Skp2 deficiency restricts the progression and stem cell features of castration-resistant prostate cancer by destabilizing Twist. *Oncogene*. 2017;36:4299–310. <https://doi.org/10.1038/ncr.2017.64>.
34. Babbio F, Pistore C, Curti L, Castiglioni I, Kunderfranco P, Brino L, Oudet P, Seiler R, Thalman GN, Roggero E, Sarti M, Pinton S, Mello-Grand M, Chiorino G, Catapano CV, Carbone GM, Bonapace IM. The SRA protein UHRF1 promotes epigenetic cross-talks and is involved in prostate cancer progression. *Oncogene*. 2012;31:4878–87. <https://doi.org/10.1038/ncr.2011.641>.
35. Ashraf W, Ibrahim A, Alhosin M, Zayter L, Ouararhni K, Papin C, Ahmad T, Hamiche A, Mély Y, Bronner C, Mousli M. The epigenetic integrator UHRF1: on the road to become a universal biomarker for cancer. *Oncotarget*. 2017;8:51946–62. <https://doi.org/10.18632/oncotarget.17393>.

36. Archer Goode E, Wang N, Munkley J. Prostate cancer bone metastases biology and clinical management. *Oncol Lett*. 2023;25:163. <https://doi.org/10.3892/ol.2023.13749>.
37. Robinson T, Escara-Wilke J, Dai J, Zimmermann J, Keller ET. A CXCR4 inhibitor (balixafortide) enhances docetaxel-mediated antitumor activity in a murine model of prostate cancer bone metastasis. *Prostate*. 2023;83:1247–54. <https://doi.org/10.1002/pros.24584>.
38. Cappell SD, Mark KG, Garbett D, Pack LR, Rape M, Meyer T. Emi1 switches from being a substrate to an inhibitor of APC/CCDH1 to start the cell cycle. *Nature*. 2018;558:313–7. <https://doi.org/10.1038/s41586-018-0199-7>.
39. Di Fiore B, Pines J. Emi1 is needed to couple DNA replication with mitosis but does not regulate activation of the mitotic APC/C. *J Cell Biol*. 2007;177:425–37. <https://doi.org/10.1083/jcb.200611166>.
40. Wang W, Kirschner MW. Emi1 preferentially inhibits ubiquitin chain elongation by the anaphase-promoting complex. *Nat Cell Biol*. 2013;15:797–806. <https://doi.org/10.1038/ncb2755>.
41. Miller JJ, Summers MK, Hansen DV, Nachury MV, Lehman NL, Loktev A, Jackson PK. Emi1 stably binds and inhibits the anaphase-promoting complex/cyclosome as a pseudosubstrate inhibitor. *Genes Dev*. 2006;20:2410–20. <https://doi.org/10.1101/gad.1454006>.
42. Machida YJ, Dutta A. The APC/C inhibitor, Emi1, is essential for prevention of rereplication. *Genes Dev*. 2007;21:184–94. <https://doi.org/10.1101/gad.1495007>.
43. Frye JJ, Brown NG, Petzold G, Watson ER, Grace CRR, Nourse A, Jarvis MA, Kriwacki RW, Peters J-M, Stark H, Schulman BA. Electron microscopy structure of human APC/C(CDH1)-Emi1 reveals multimodal mechanism of E3 ligase shutdown. *Nat Struct Mol Biol*. 2013;20:827–35. <https://doi.org/10.1038/nsmb.2593>.
44. Shimizu N, Nakajima NI, Tsunematsu T, Ogawa I, Kawai H, Hirayama R, Fujimori A, Yamada A, Okayasu R, Ishimaru N, Takata T, Kudo Y. Selective enhancing effect of early mitotic inhibitor 1 (Emi1) depletion on the sensitivity of doxorubicin or X-ray treatment in human cancer cells. *J Biol Chem*. 2013;288:17238–52. <https://doi.org/10.1074/jbc.M112.446351>.
45. Gong J, Zhou Y, Liu D, Huo J. F-box proteins involved in cancer-associated drug resistance. *Oncol Lett*. 2018;15:8891–900. <https://doi.org/10.3892/ol.2018.8500>.
46. Saha D, Dang HX, Zhang M, Quigley DA, Feng FY, Maher CA. Single cell-transcriptomic analysis informs the lncRNA landscape in metastatic castration resistant prostate cancer. *Npj Genom Med*. 2024;9:14. <https://doi.org/10.1038/s41525-024-00401-3>.
47. Wang Z, Liu P, Inuzuka H, Wei W. Roles of F-box proteins in cancer. *Nat Rev Cancer*. 2014;14:233–47. <https://doi.org/10.1038/nrc3700>.
48. Yan L, Lin M, Pan S, Assaraf YG, Wang Z-W, Zhu X. Emerging roles of F-box proteins in cancer drug resistance. *Drug Resist Updat*. 2020;49: 100673. <https://doi.org/10.1016/j.drug.2019.100673>.
49. Song Y, Lin M, Liu Y, Wang Z-W, Zhu X. Emerging role of F-box proteins in the regulation of epithelial-mesenchymal transition and stem cells in human cancers. *Stem Cell Res Ther*. 2019;10:124. <https://doi.org/10.1186/s13287-019-1222-0>.
50. Wang R, Gao X, Xie L, Lin J, Ren Y. METTL16 regulates the mRNA stability of FBXO5 via m6A modification to facilitate the malignant behavior of breast cancer. *Cancer Metab*. 2024;12:22. <https://doi.org/10.1186/s40170-024-00351-5>.
51. Liu P, Wang X, Pan L, Han B, He Z. Prognostic significance and immunological role of FBXO5 in human cancers: a systematic pan-cancer analysis. *Front Immunol*. 2022;13: 901784. <https://doi.org/10.3389/fimmu.2022.901784>.
52. Ji J, Jing A, Ding Y, Ma X, Qian Q, Geng T, Cheng W, Zhang M, Sun Q, Ma S, Wang X, Yuan Q, Xu M, Qin J, Ma L, Yang J, He J, Du Q, Xia M, Xu Y, Chen Z, Zhu L, Liu W, Liu S, Liu B. FBXO5-mediated RNF183 degradation prevents endoplasmic reticulum stress-induced apoptosis and promotes colon cancer progression. *Cell Death Dis*. 2024;15:33. <https://doi.org/10.1038/s41419-024-06421-2>.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.